

Common Architectures in Deep Learning

Overview of widely used deep learning models

Feedforward Neural Networks (FNN)

- • Simplest type of neural network with unidirectional data flow.
- • Used for basic classification and regression tasks.
- • Layers: Input → Hidden Layers → Output.

Convolutional Neural Networks (CNN)

- • Designed for image/video data using convolutional layers.
- • Key components: Convolution, Pooling, Fully Connected layers.
- • Applications: Image classification, object detection.

Recurrent Neural Networks (RNN)

- • Maintains hidden states for sequential data.
- • Applications: Time series forecasting, NLP, speech recognition.
- • Limitation: Vanishing gradient for long sequences.

LSTM and GRU Networks

- • LSTM: Uses gates (input, forget, output) to capture long-term dependencies.
- • GRU: Simplified version of LSTM with fewer gates, faster training.
- • Applications: Language modeling, text generation, sentiment analysis.

GANs and Autoencoders

- • GANs: Generator vs. Discriminator for realistic data generation.
- • Autoencoders: Compress and reconstruct data, used for anomaly detection.
- • Applications: Image synthesis, denoising, style transfer.

Transformer and ResNet

- • Transformers: Self-attention for NLP, widely used in models like BERT, GPT.
- • ResNet: Skip connections to solve vanishing gradient in deep networks.
- • Applications: Text summarization, image classification.

Capsule Networks (CapsNet)

- Captures spatial relationships using capsules.
- More robust to object orientation and pose.
- Applications: Advanced image recognition.

Summary

- • Deep learning architectures vary based on data type and task.
- • CNNs dominate computer vision, while RNNs/Transformers dominate NLP.
- • GANs, Autoencoders, and ResNet represent advanced innovations.
- • Choosing the right architecture is key to performance.